

FUSE – Synligare Seminar

III Tool Example System Development - Synligare

May 22 2015

Legend

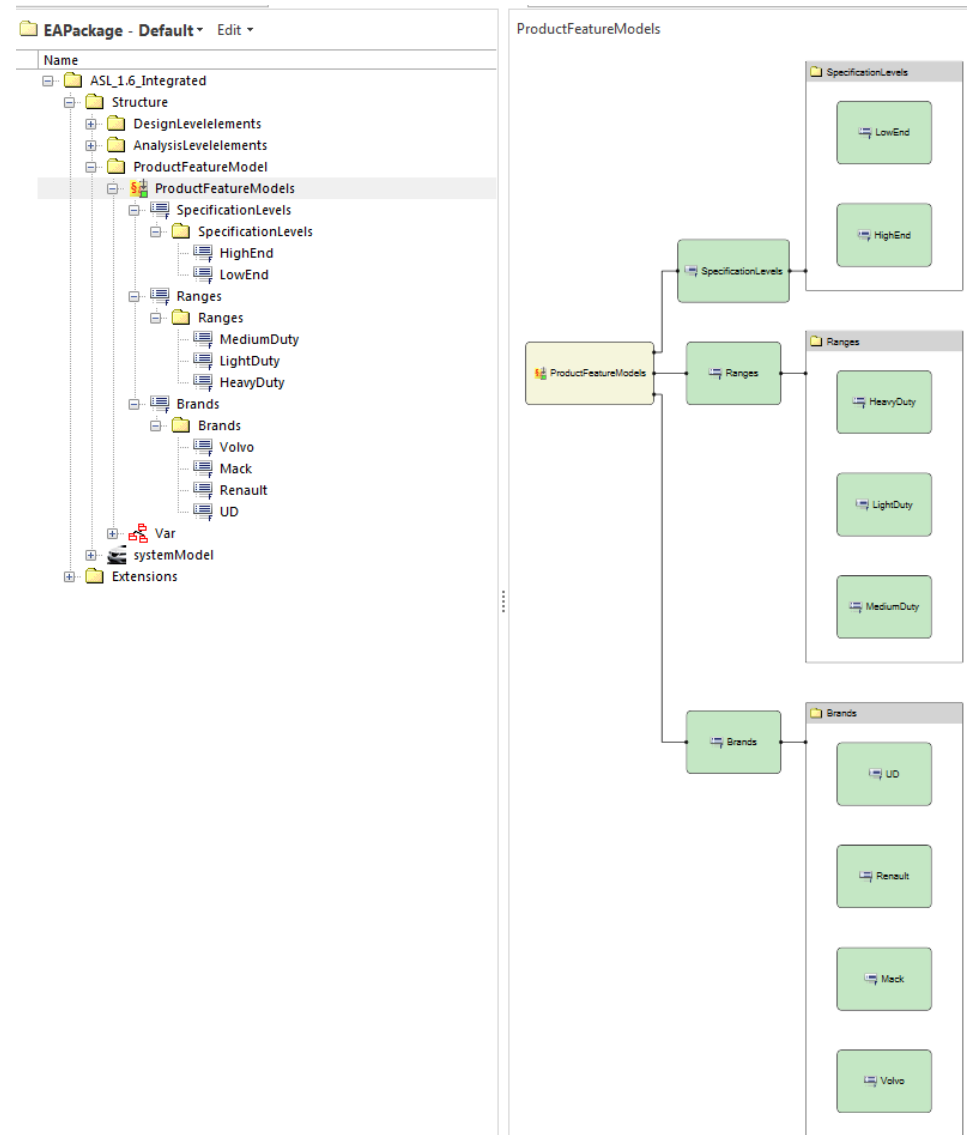
 OEM Activities

 Tier1 Activities



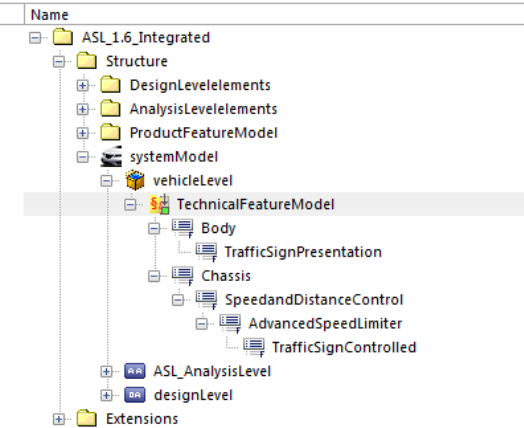
Define

Product features



Technical features

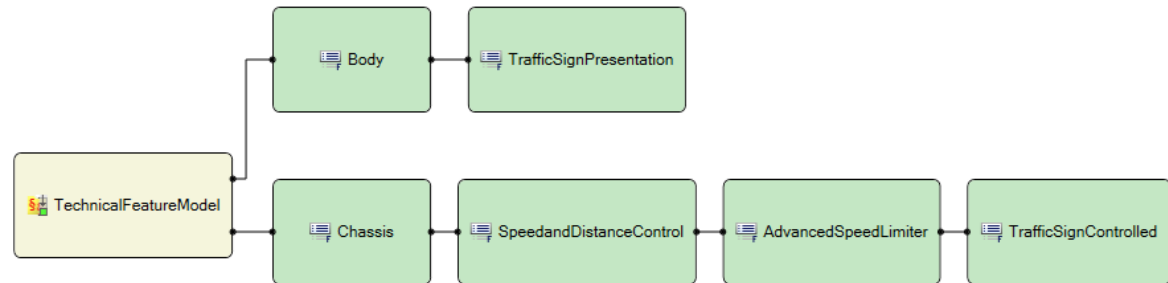
EAPackage - Default Edit



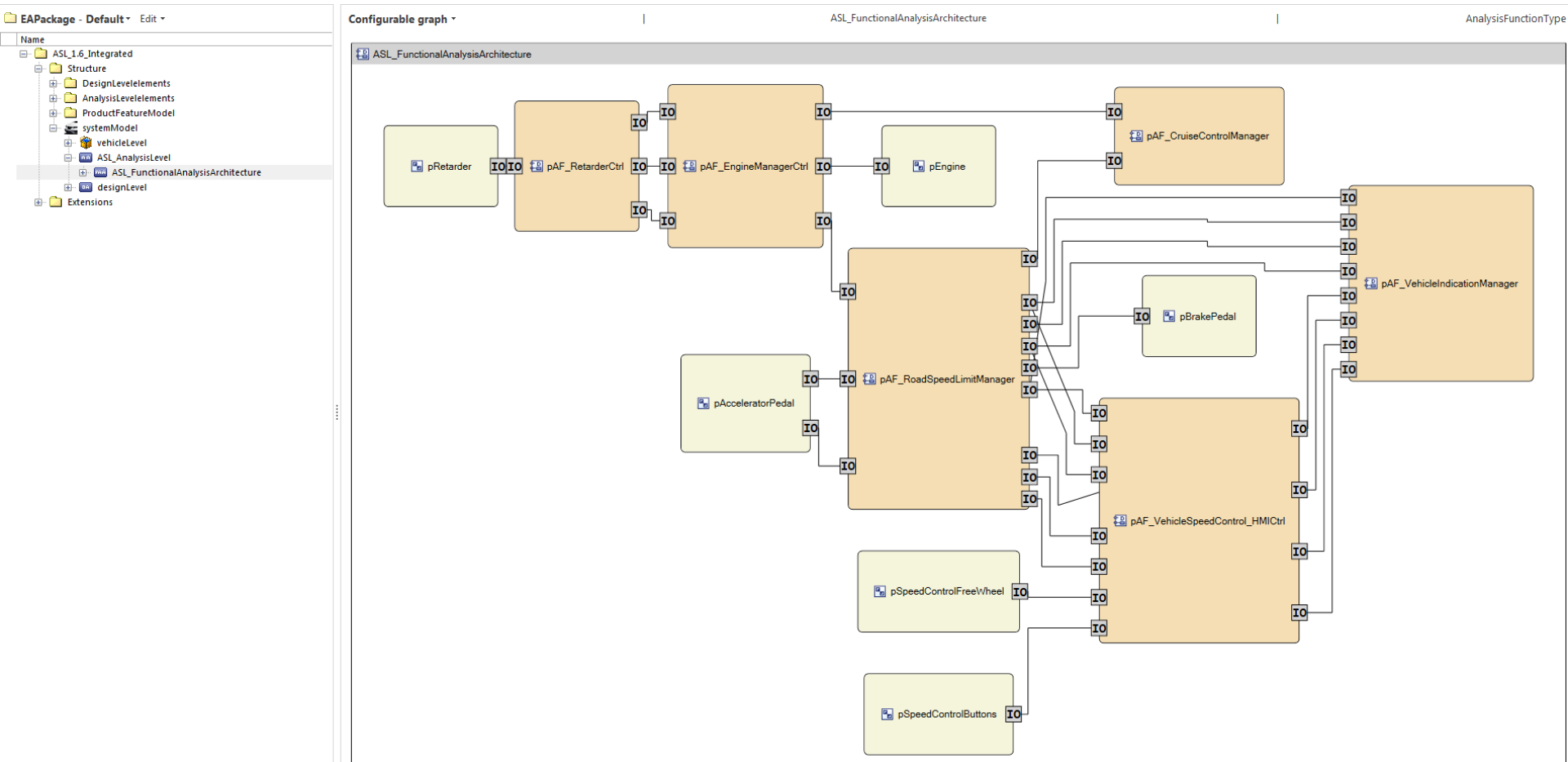
Configurable graph

TechnicalFeatureModel

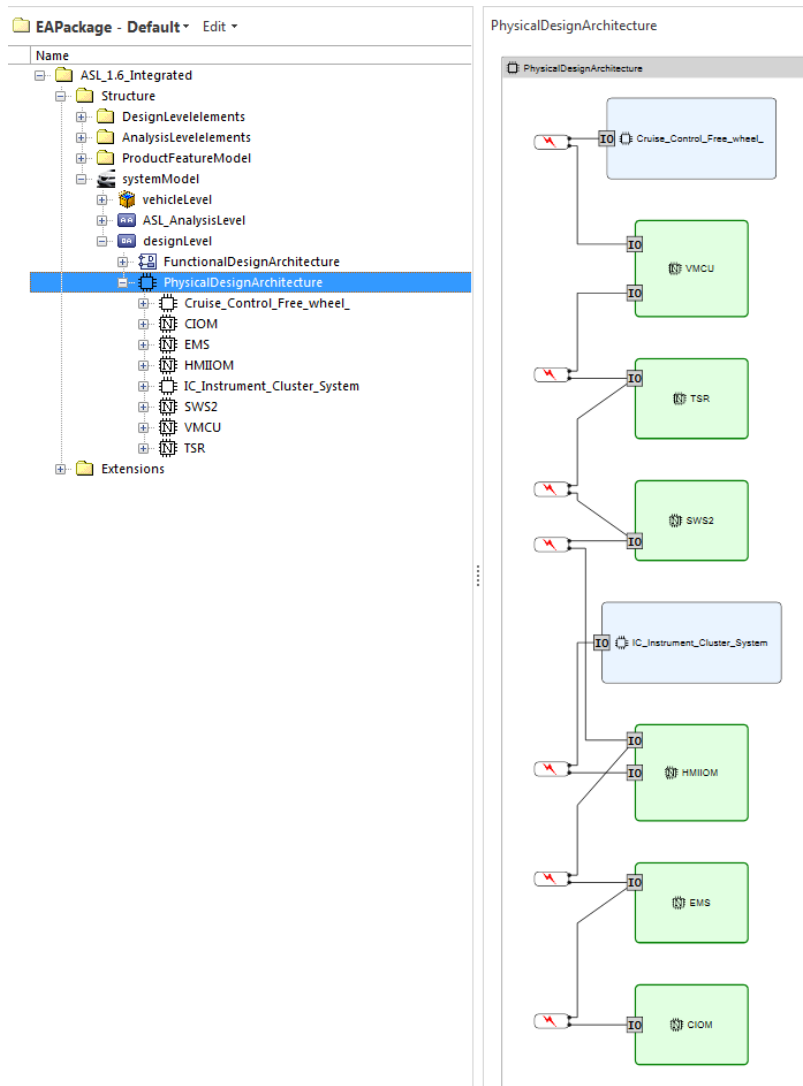
FeatureModel



Analysis level



Physical design

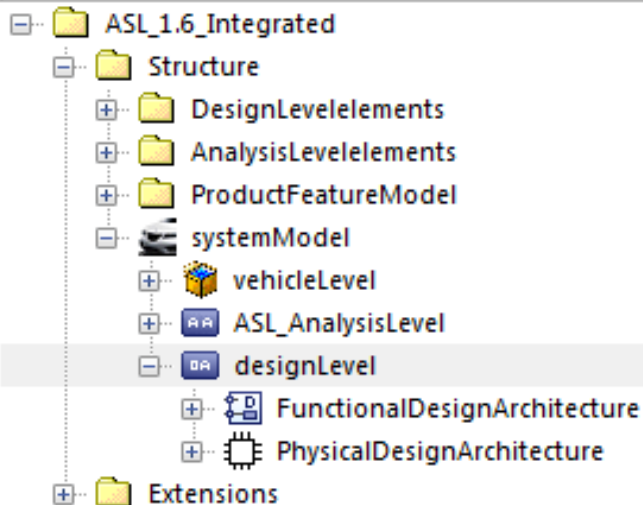




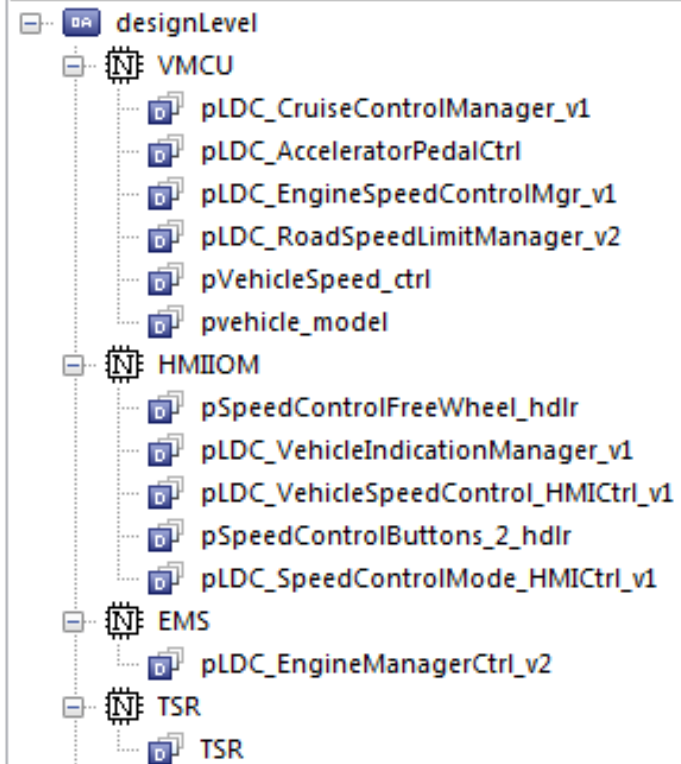
Allocation

EAPackage - Default(modified) ▾ Edit ▾

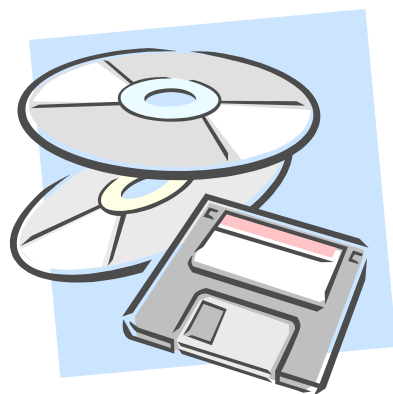
Name



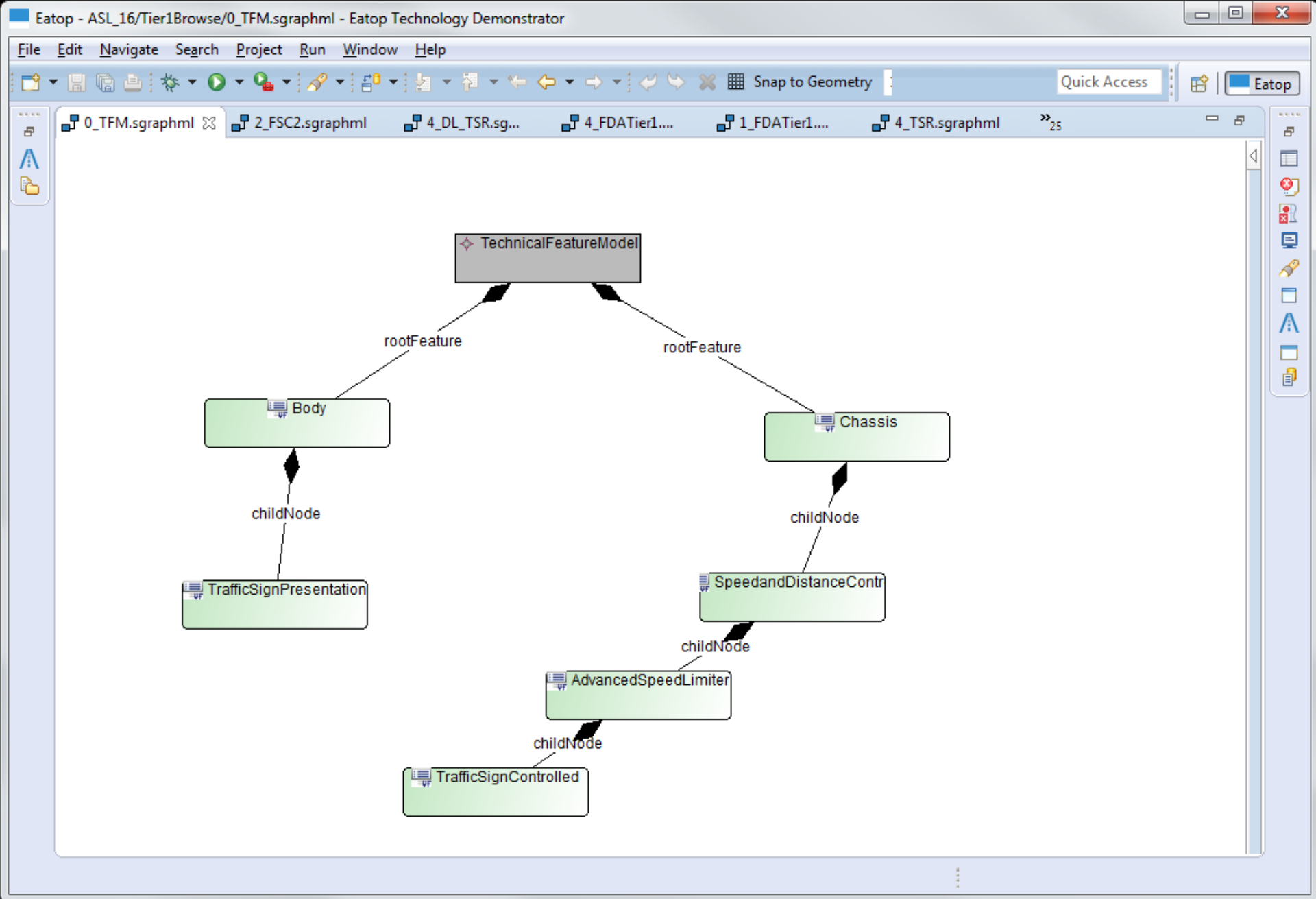
Allocation ▾



Export



Browse



TSR_A1

3_EMFCompar...

3_TSR.sgraphml

4_TSRWhite....

4_Requiremen...

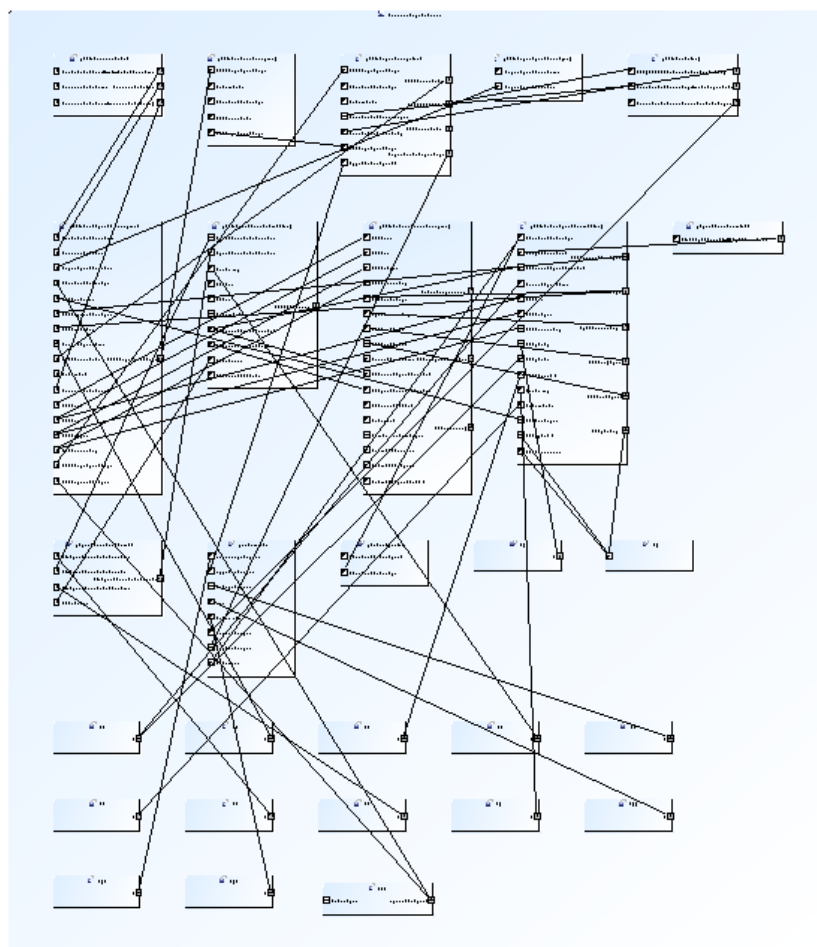
*ACC_V1

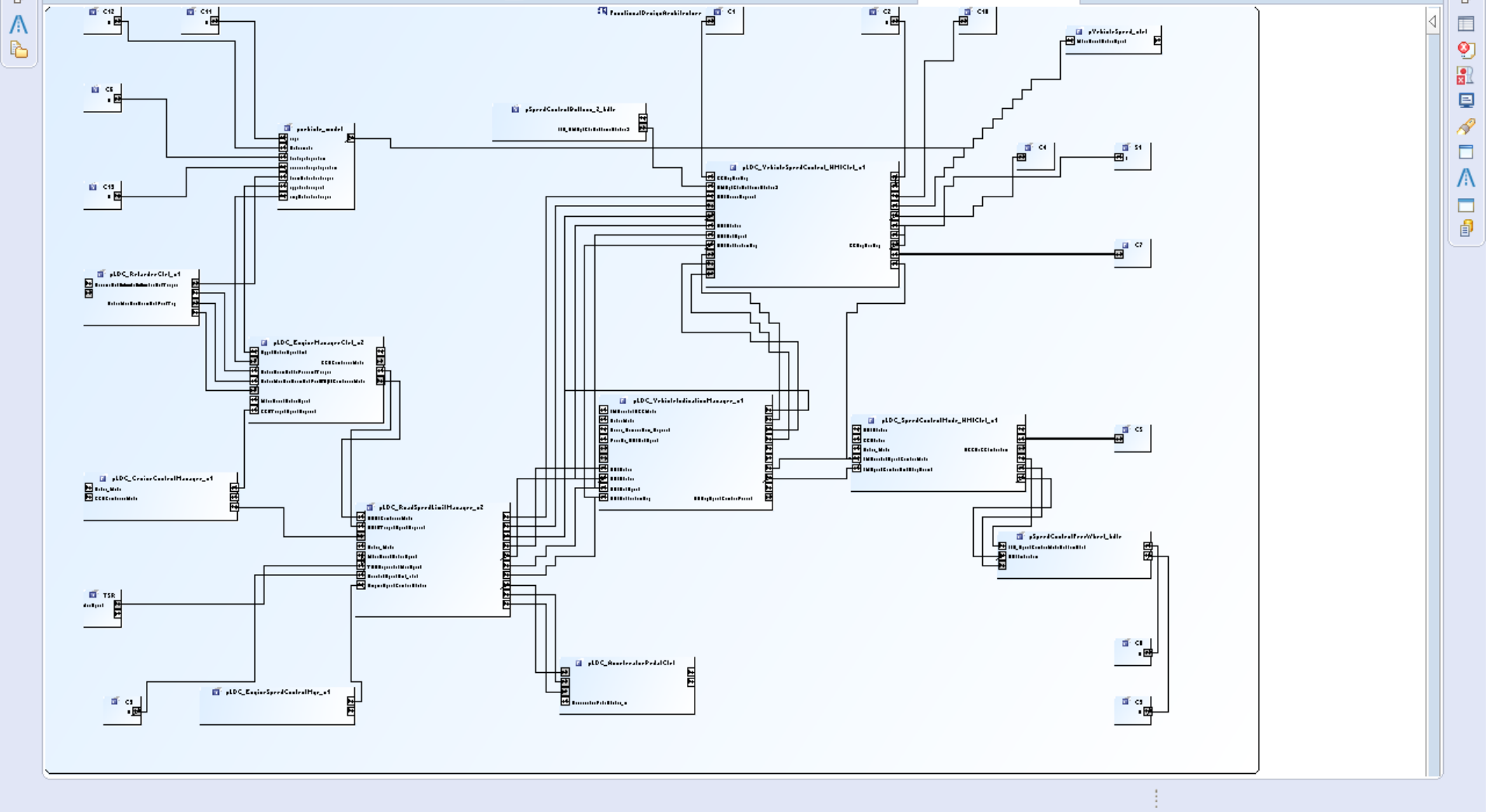
8_FSC_4.sgr...

1_FDA_OEM.s...

*2_FDA_OEMr...

>>22





Compare ('EAXML' - 'EAXML')

Model differences (2 over 2 differences still to be merged — 0 differences filtered from view)

EAXML

Extensions

RequirementsPackage

AnalysisLevelRequirements

TSR_A1

The traffic sign speed shall be updated when a ... [text changed]

TSR_A4 [requirement add]

Model Compare (Text Feature)

1ed shall be updated when a new traffic speed sign is detected and classified.

1lc sign speed shall be updated when a new traffic speed sign is detected.

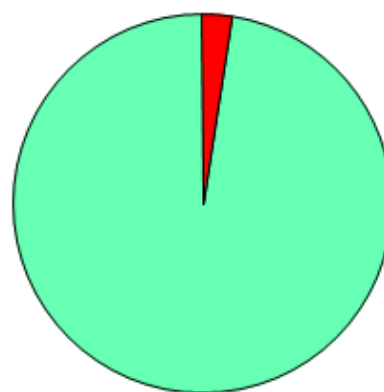
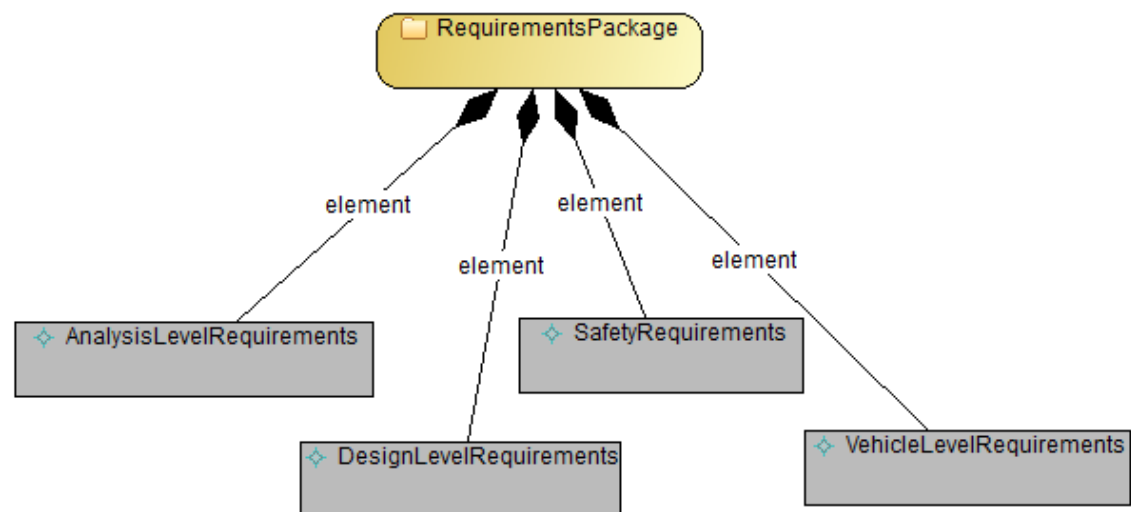
Left: 1 : 1, Right: 1 : 1, incoming change #1 (Left: 1 : 1, Right: 1 : 1)

?

Commit

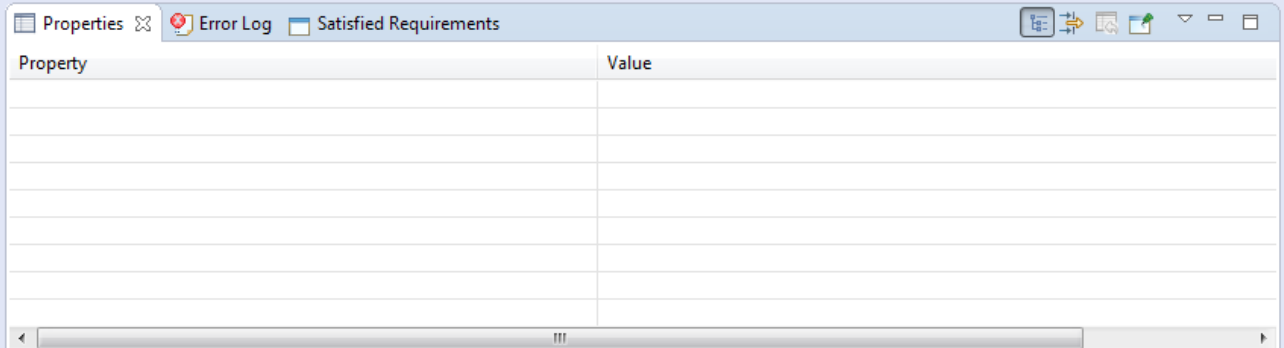
Cancel

- main
 - 4_Requirements.sgraphml
 - ASL_1.6_OEM.eaxml
 - EAXML [EAXML]
 - EASTADLExtensionElements [EAPackage]
 - Extensions [EAPackage]
 - DependabilityPackage [EAPackage]
 - EnvironmentPackage [EAPackage]
 - RequirementsPackage [EAPackage]
 - RequirementsModels [4 items]
 - AnalysisLevelRequirements [RequirementsModel]
 - DesignLevelRequirements [RequirementsModel]
 - SafetyRequirements [RequirementsModel]
 - VehicleLevelRequirements [RequirementsModel]
 - TakeRateAnnotations [EAPackage]
 - TimingPackage [EAPackage]
 - VerificationValidationWiP [EAPackage]
 - Structure [EAPackage]
 - AnalysisLevelElements [EAPackage]
 - DesignLevelElements [EAPackage]
 - ProductFeatureModel [EAPackage]
 - systemModel [SystemModel]



Satisfied Requirements, (323)

Not satisfied Requirements, (8)

 Project Explorer

Roundtrip Tier 1

- First step: Import Model

Modeling EAST-ADL in Enterprise Architect

- EAST-ADL profile in Enterprise Architect (but limited)
- Model format differs in Enterprise Architect / UML
- UML is view-based. This is not built in to EAST-ADL



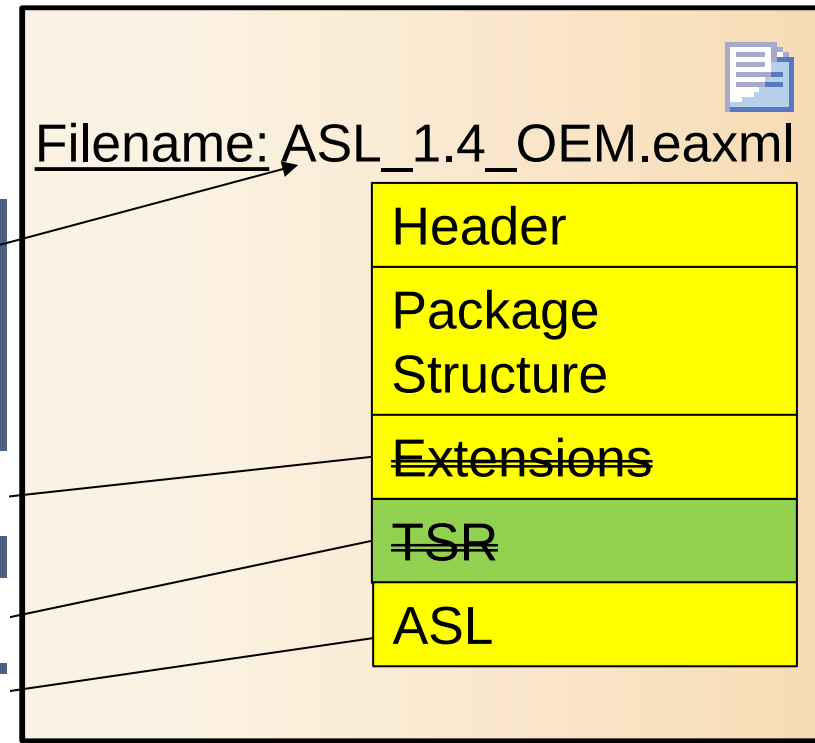
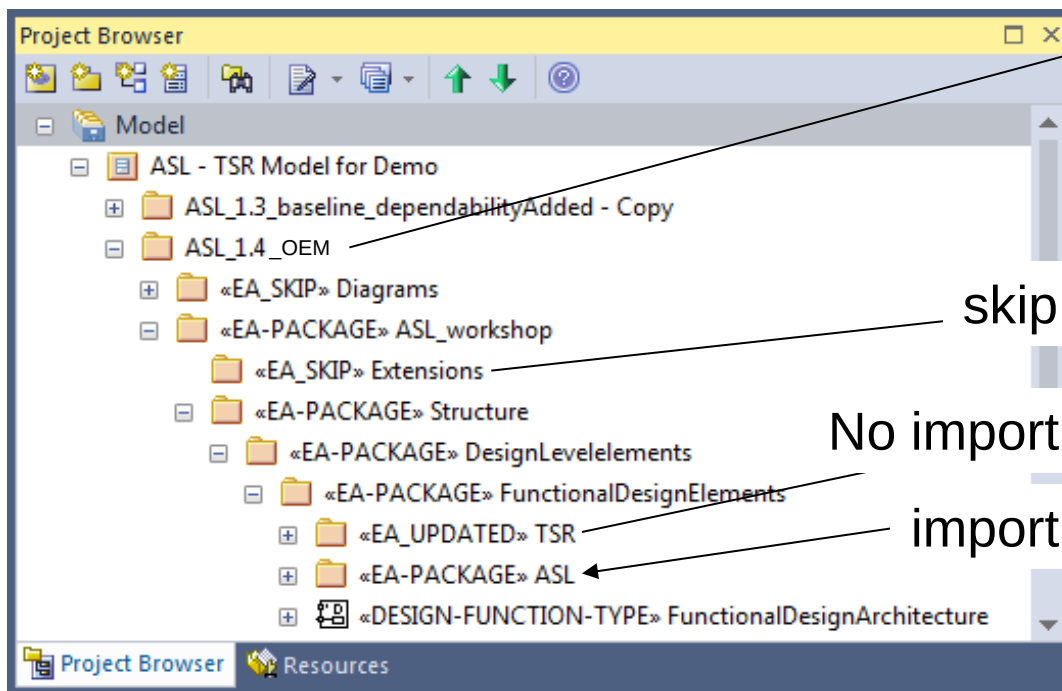
Model roundtrip principle for Enterprise Architect

- Full round-trip not supported
- Exchange format is .eaxml
- Only parts of EAST-ADL is supported
- UUID is set correctly, but during import only package path is used

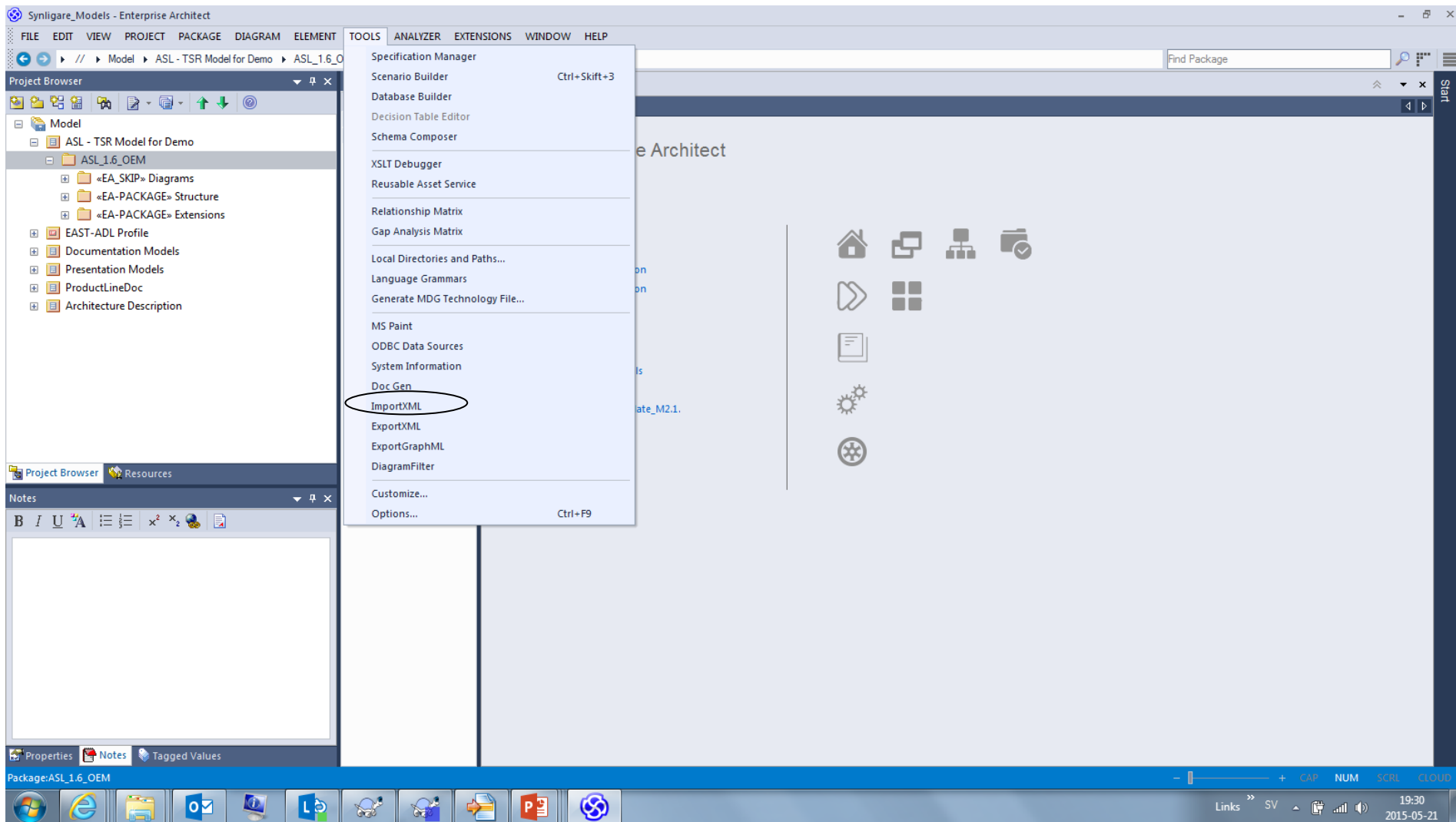


Handling of different type of models in Enterprise Architect

- Import from .eaxml-file



Tier1 – First import / Sync



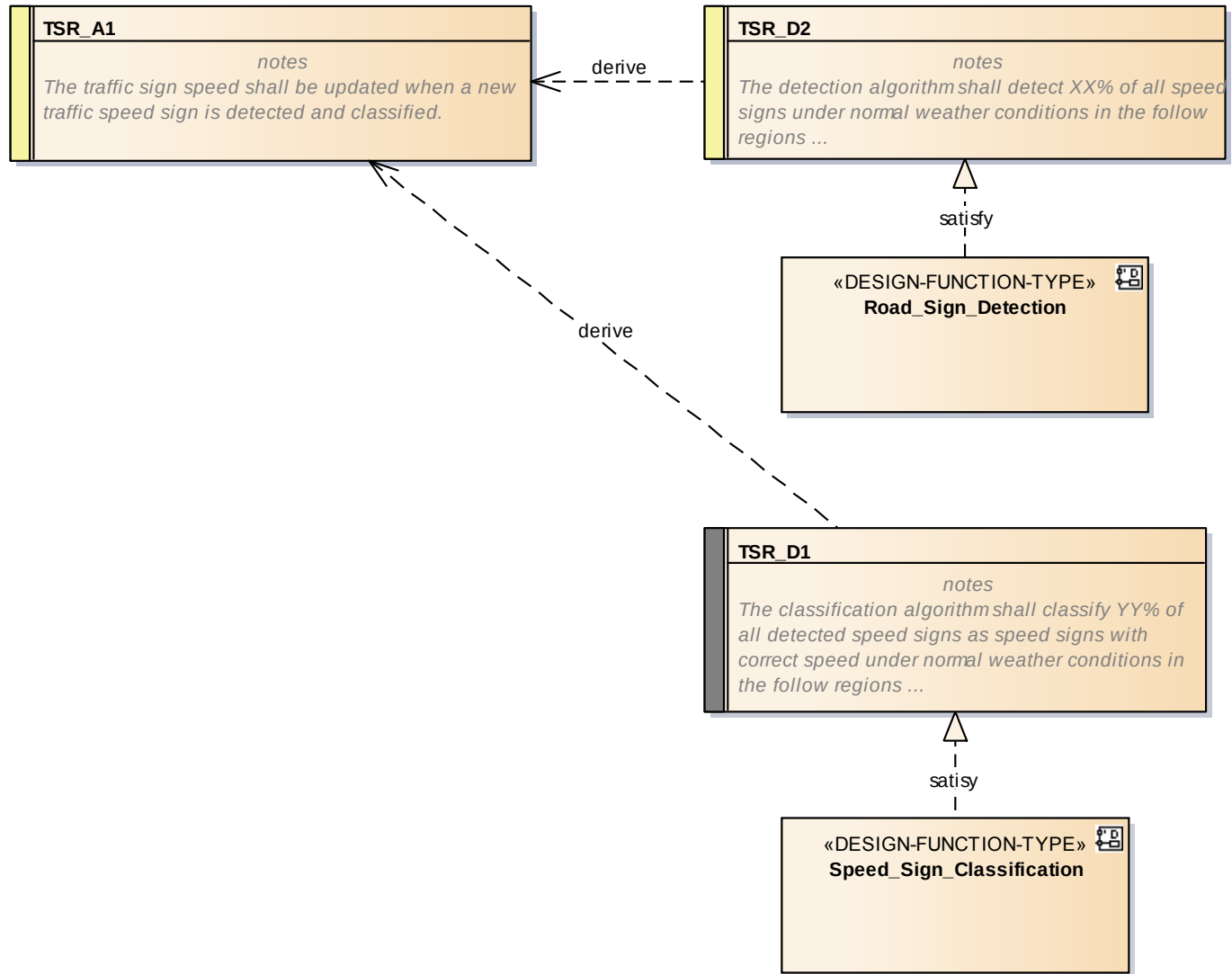
Refine

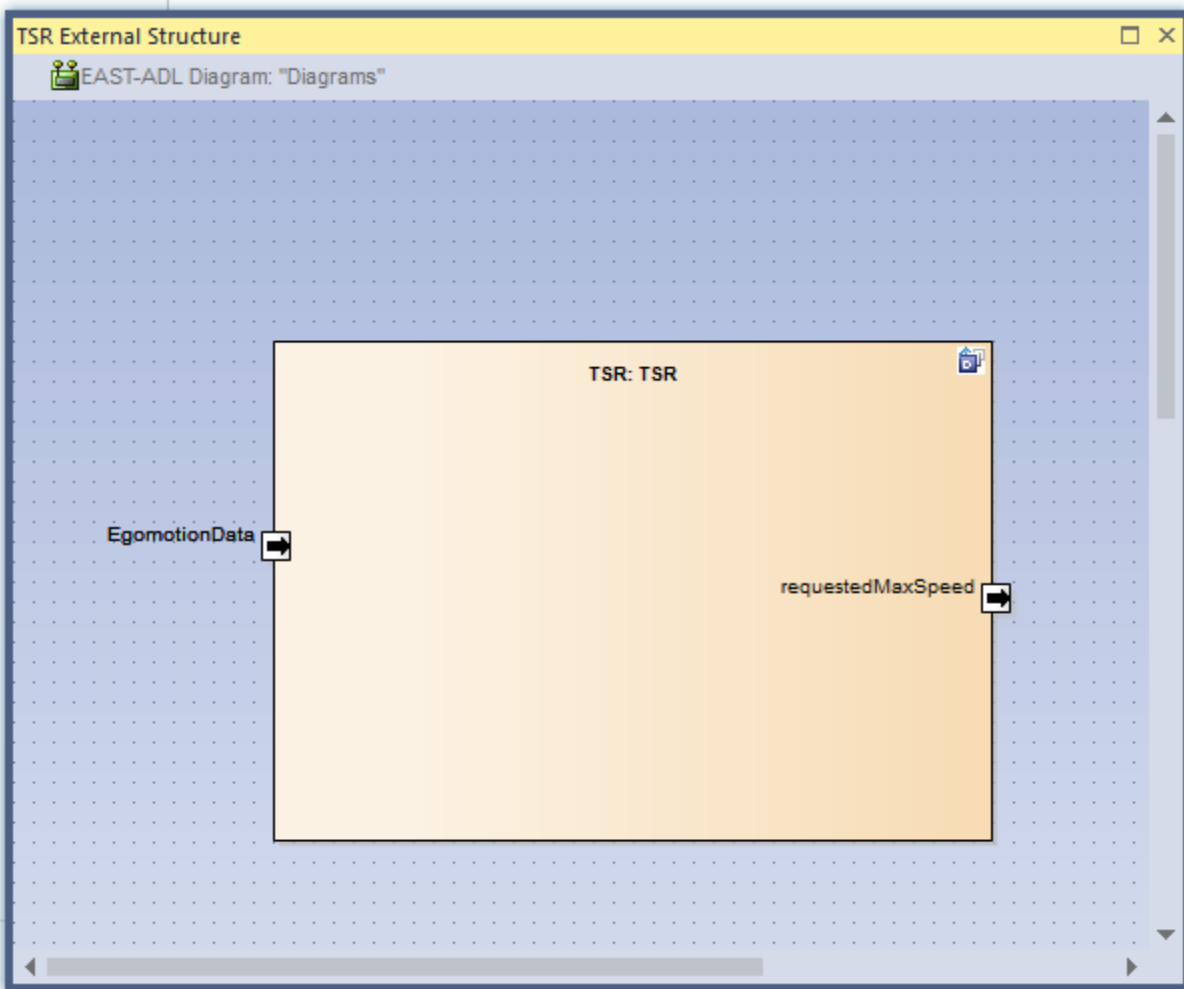
Refine

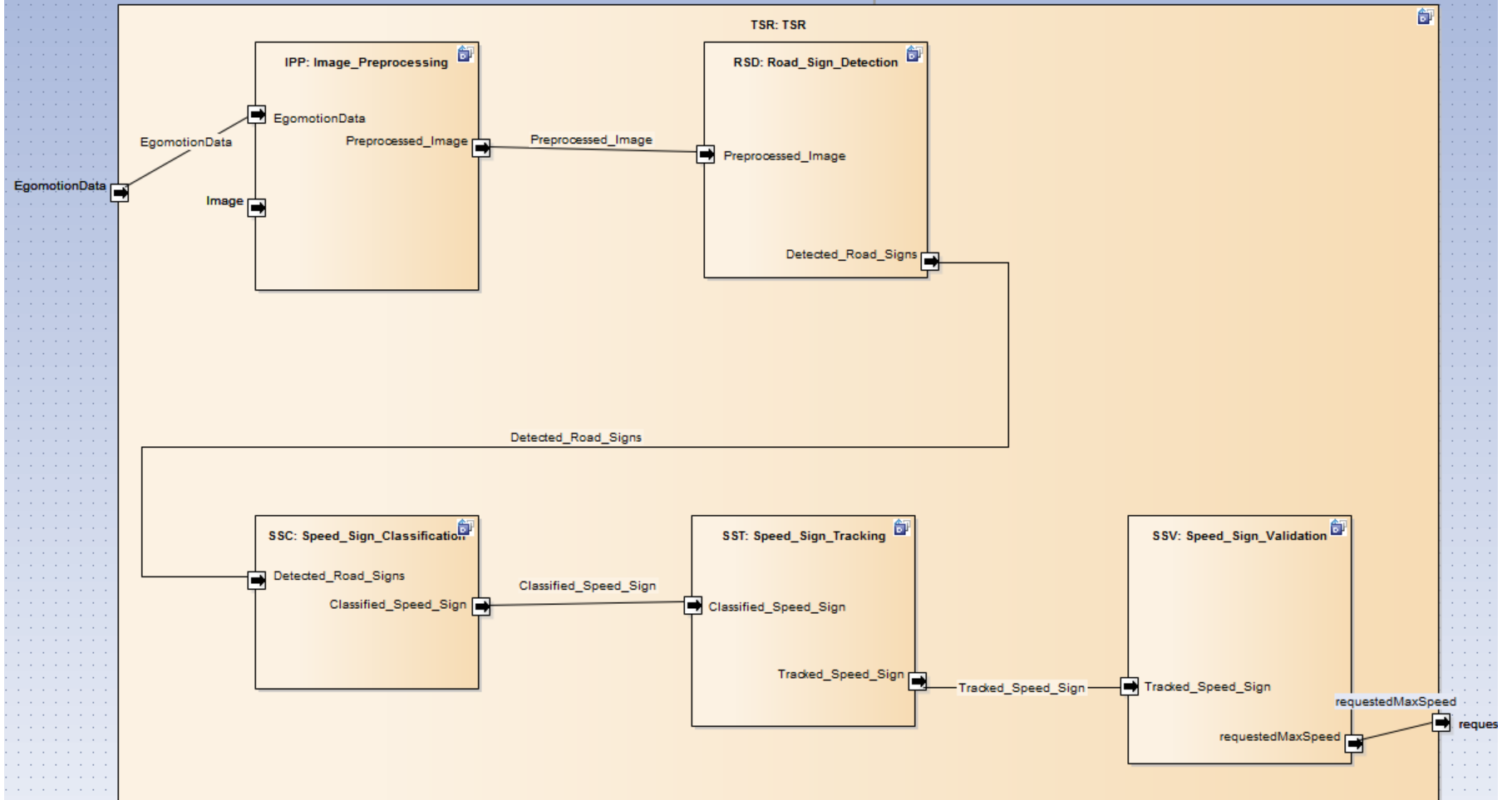
- Accept requirements
- Refine requirements
- Check Architectural Interfaces
- Refine Architecture
- Realize Requirements with design elements
- Add views and Metrics

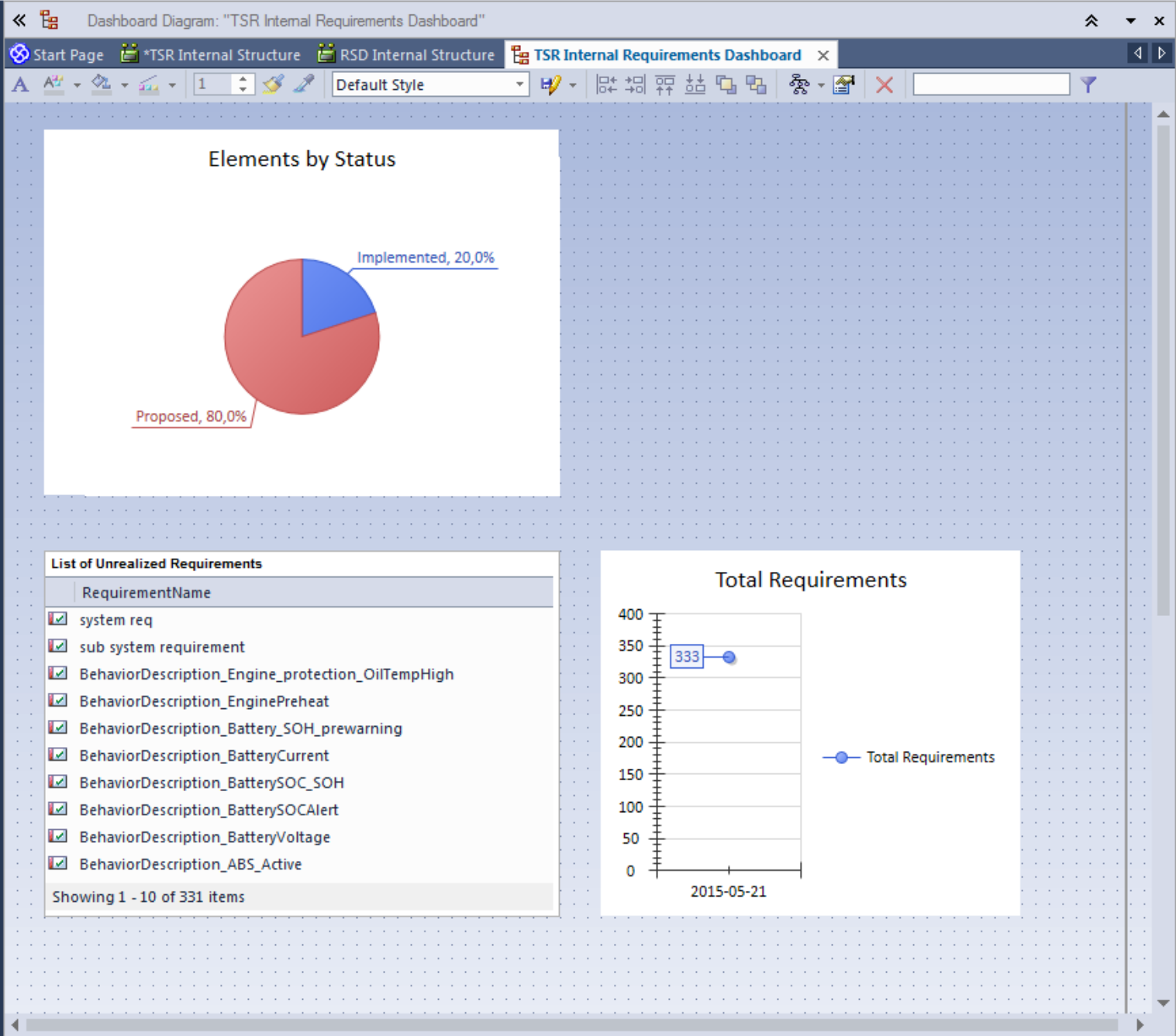


TSR Requirement trace





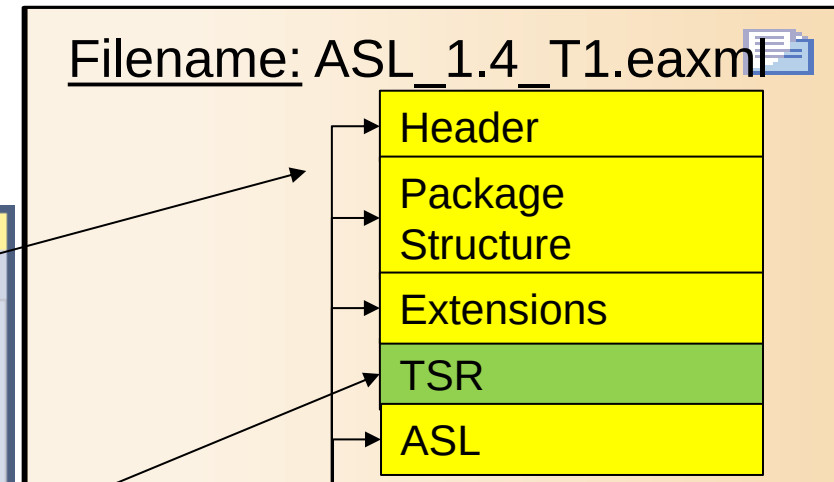
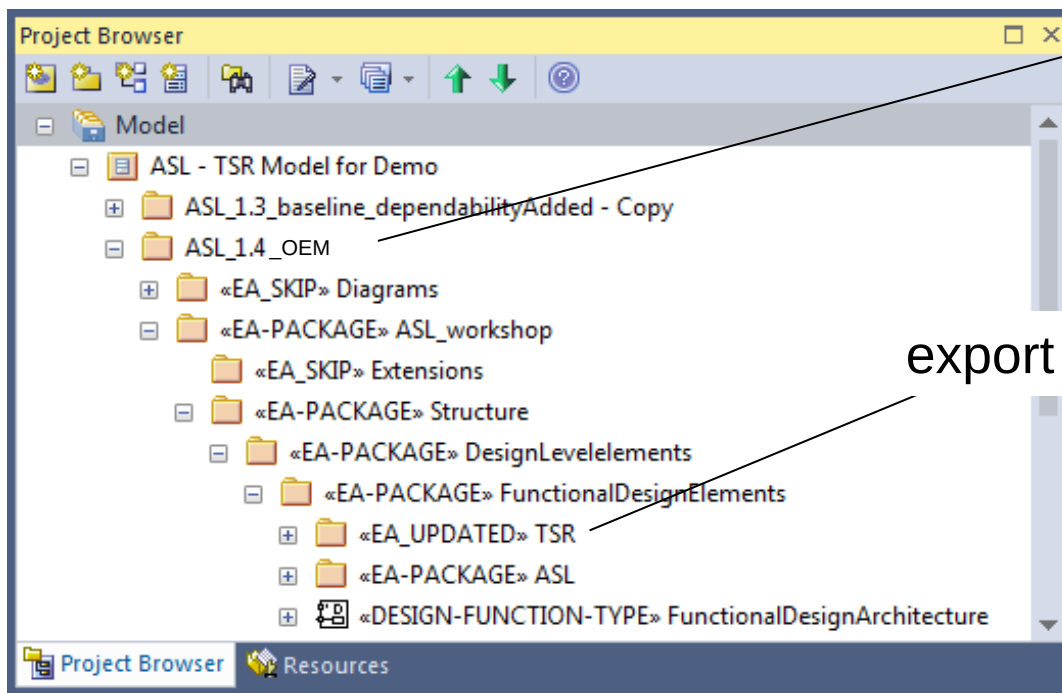




Export

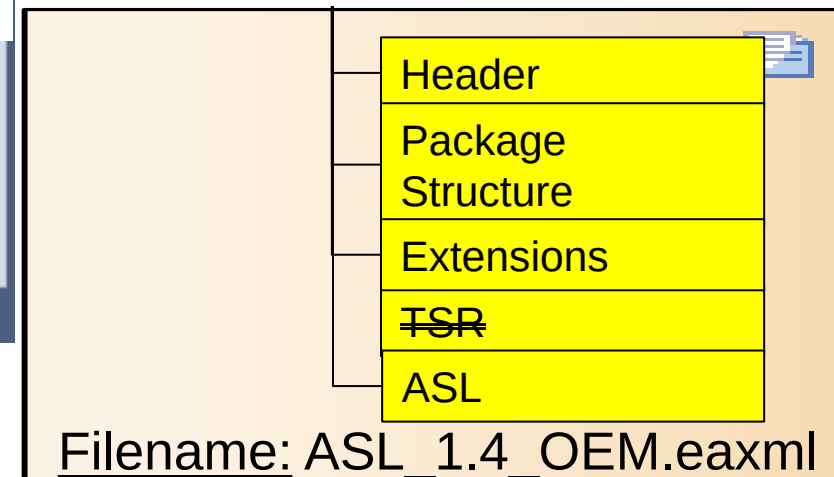
Handling of different type of models in Enterprise Architect

- Export to .eaxml-file



export

copy



Tier1 – Export model

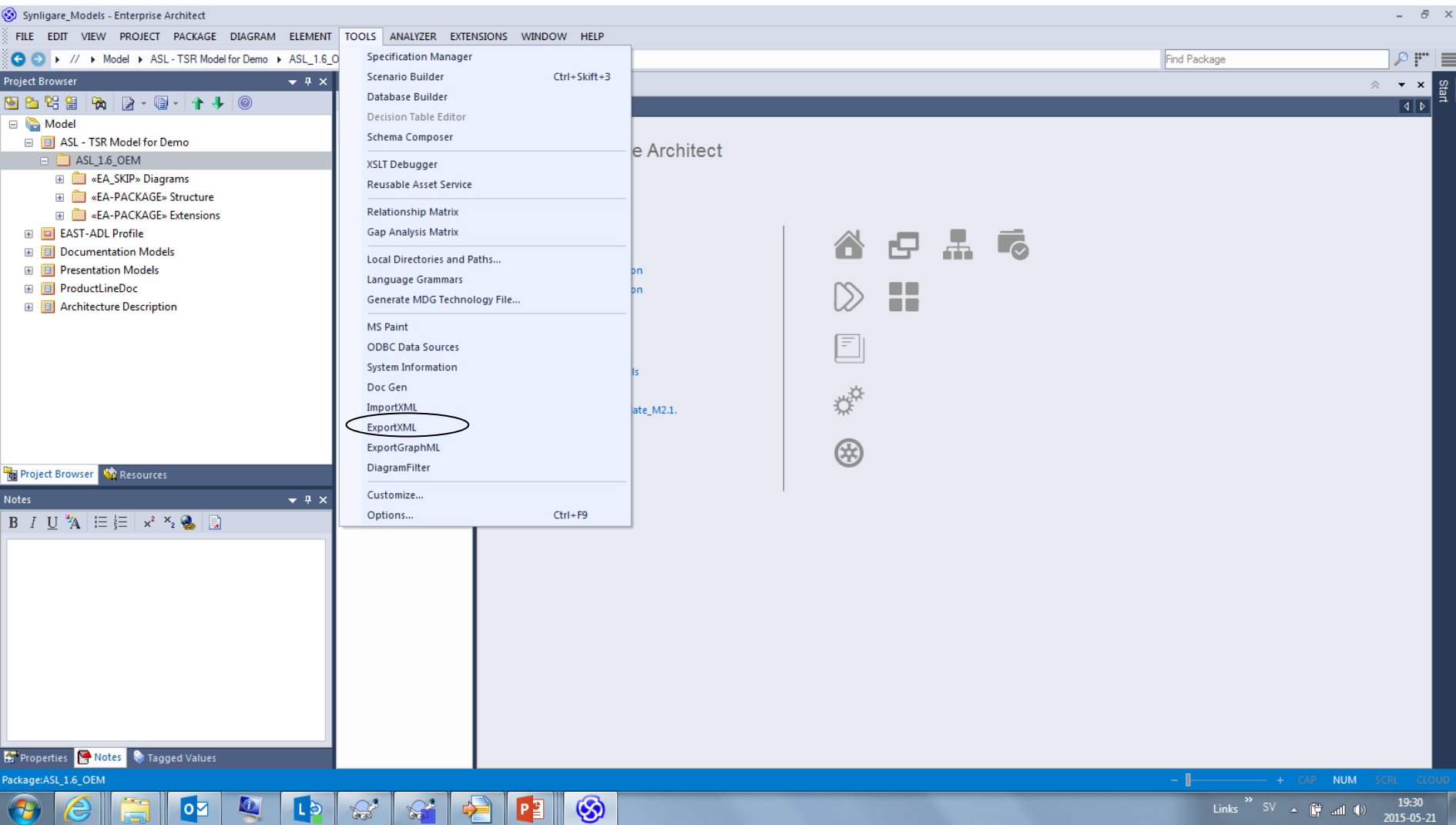
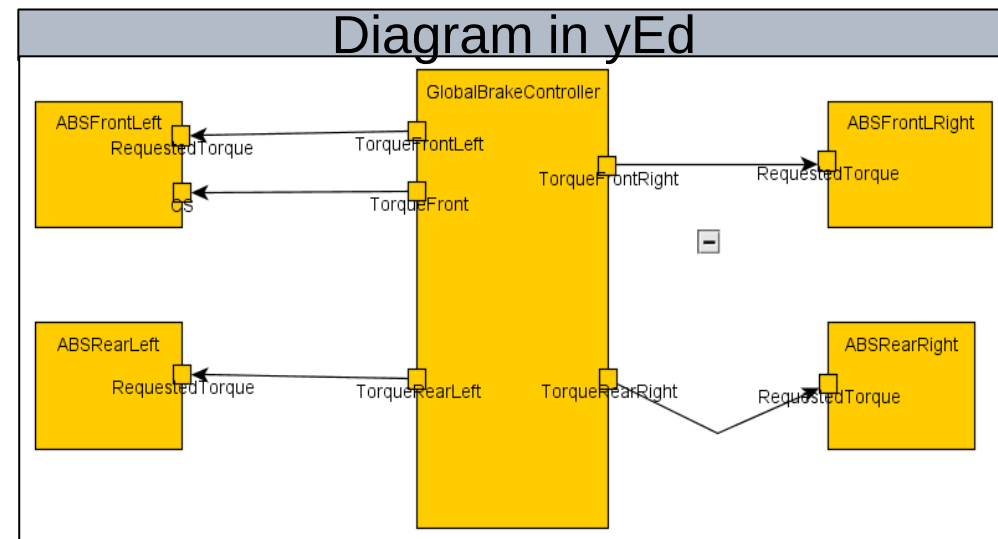
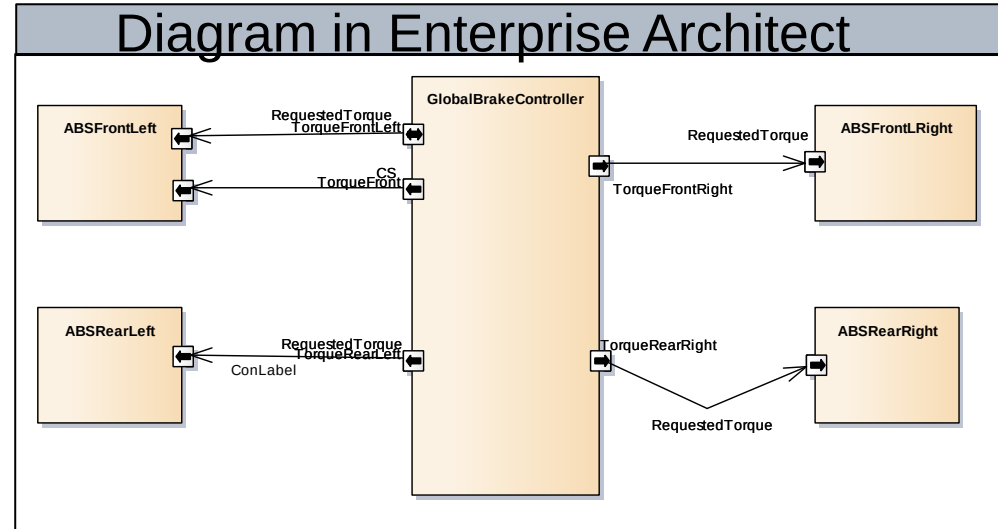


Diagram Exchange

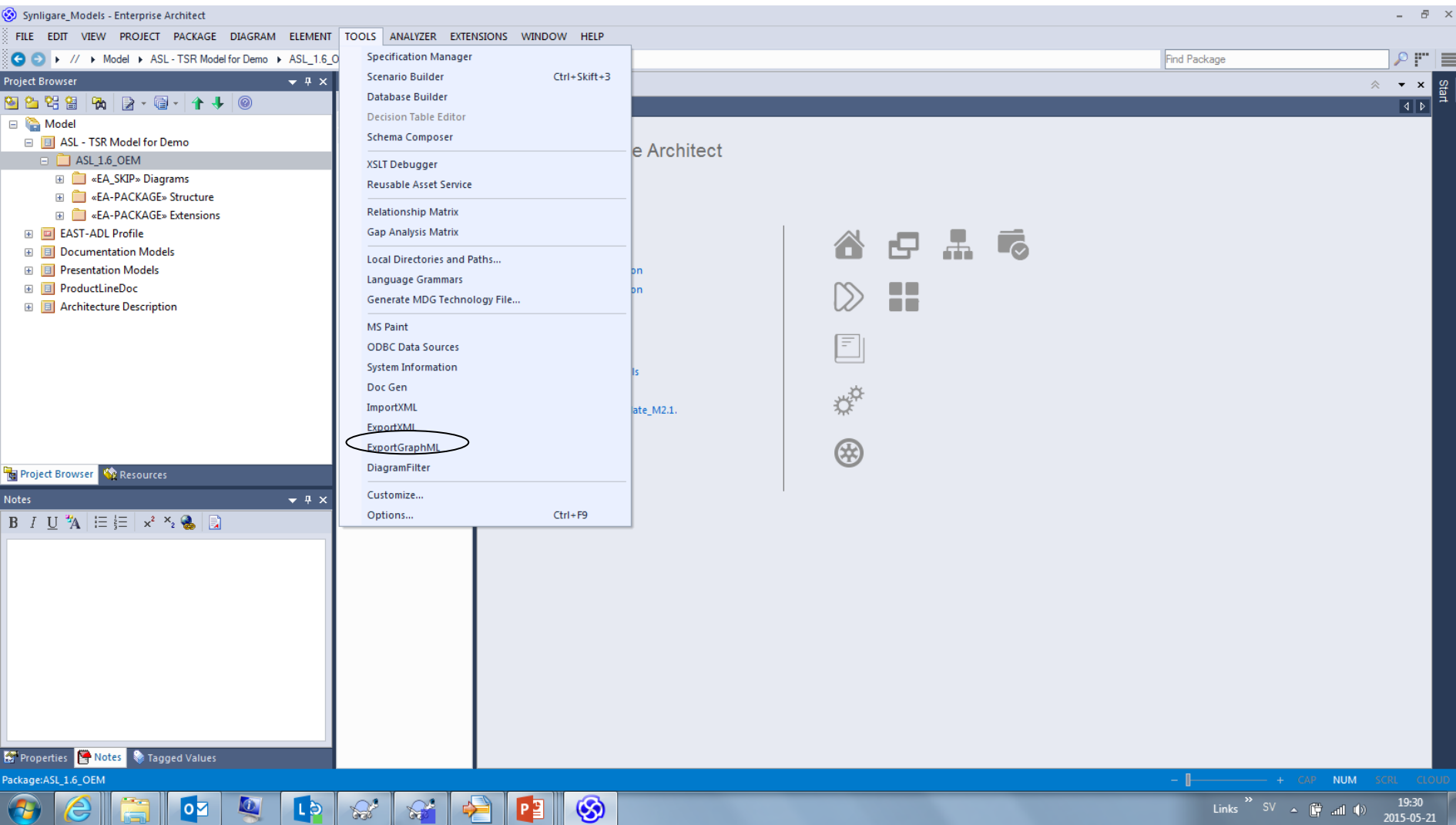
No roundtrip supported

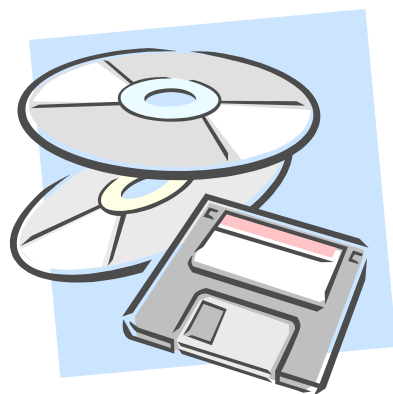
Export from EA to GraphML

- Create diagram in EA
- Export to GraphML
- Show in yEd

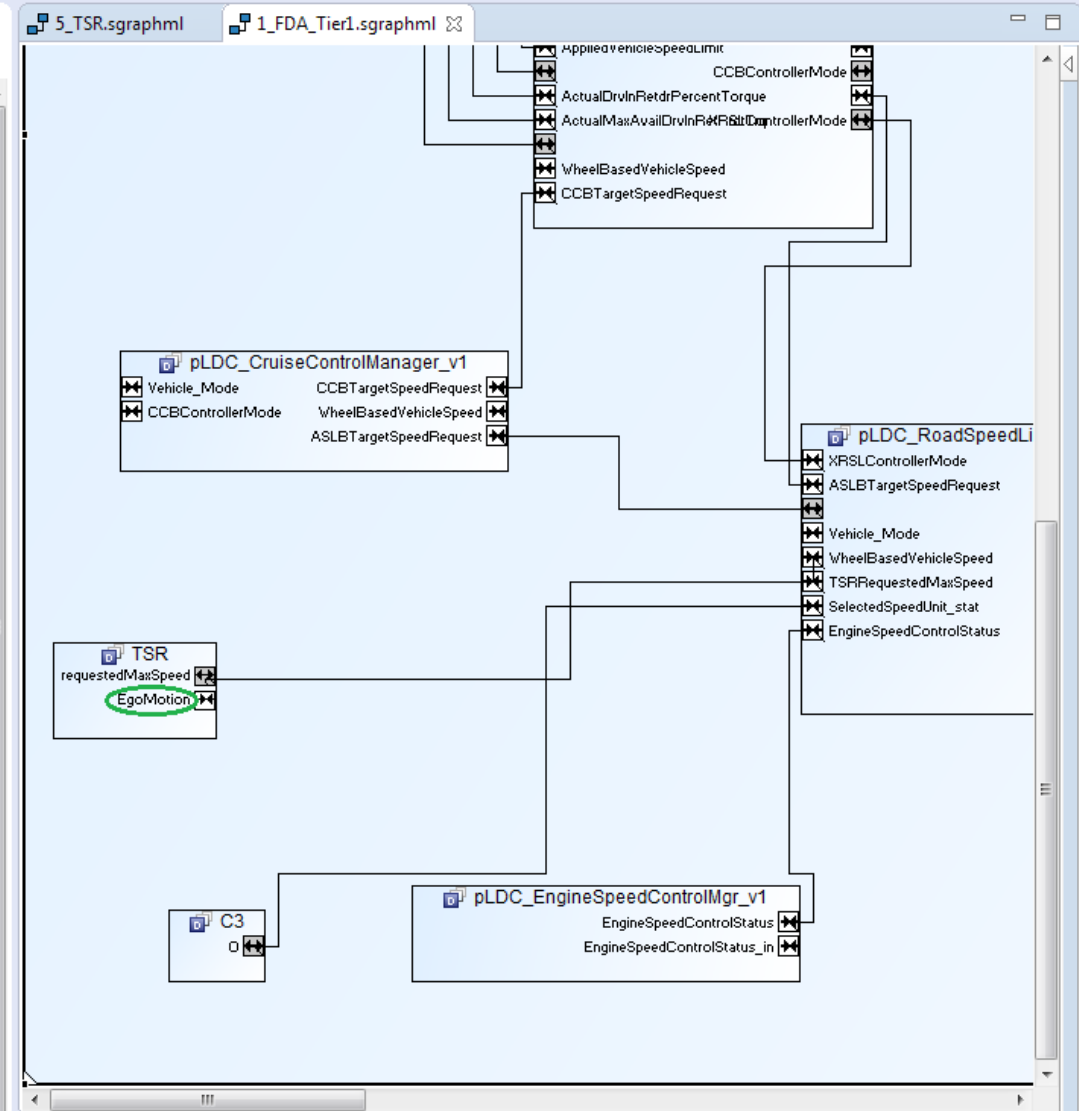


Tier1 – Export model





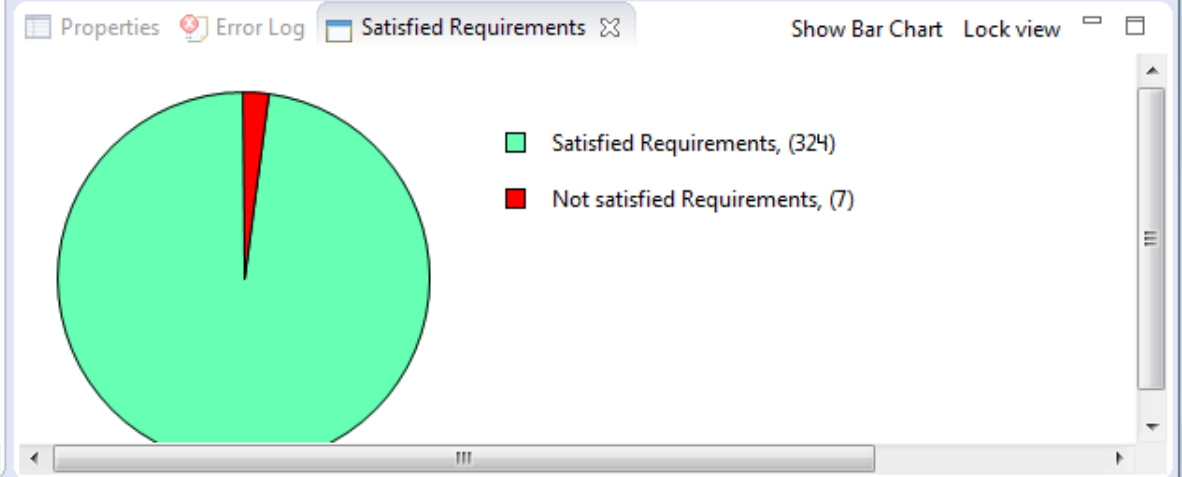
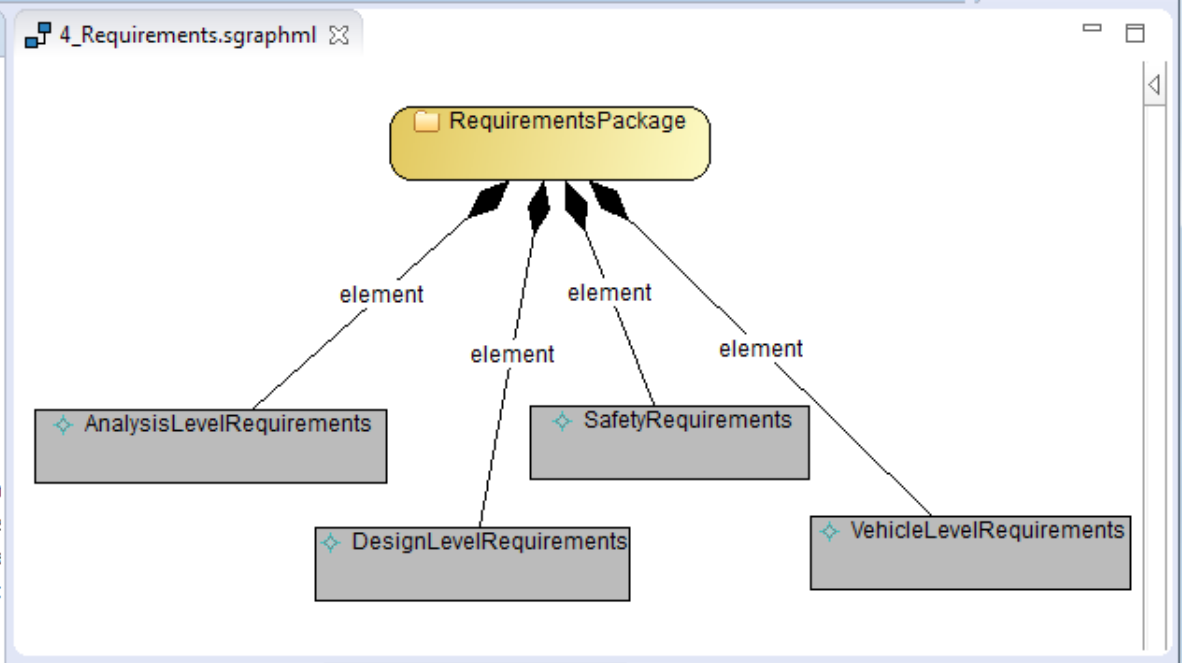
Browse



EAST-ADL Explorer

Project Explorer

- main
 - 4_Requirements.sgraphml
 - ASL_1.6_Tier1.eaxml
 - EAXML [EAXML]
 - EASTADLExtensionElements [EAPackage]
 - DependabilityPackage [EAPackage]
 - Extensions [EAPackage]
 - DependabilityPackage [EAPackage]
 - EnvironmentPackage [EAPackage]
 - RequirementsPackage [EAPackage]
 - RequirementsModels [4 items]
 - AnalysisLevelRequirements [Requirement]
 - DesignLevelRequirements [Requirement]
 - SafetyRequirements [RequirementsMode]
 - VehicleLevelRequirements [Requirement]
 - TakeRateAnnotations [EAPackage]
 - ASLTakeRate [TakeRateConstraint]
 - VehicleVolumes [TakeRateConstraint]
 - TimingPackage [EAPackage]
 - VerificationValidationWiP [EAPackage]
 - Structure [EAPackage]





§ 4 items selected

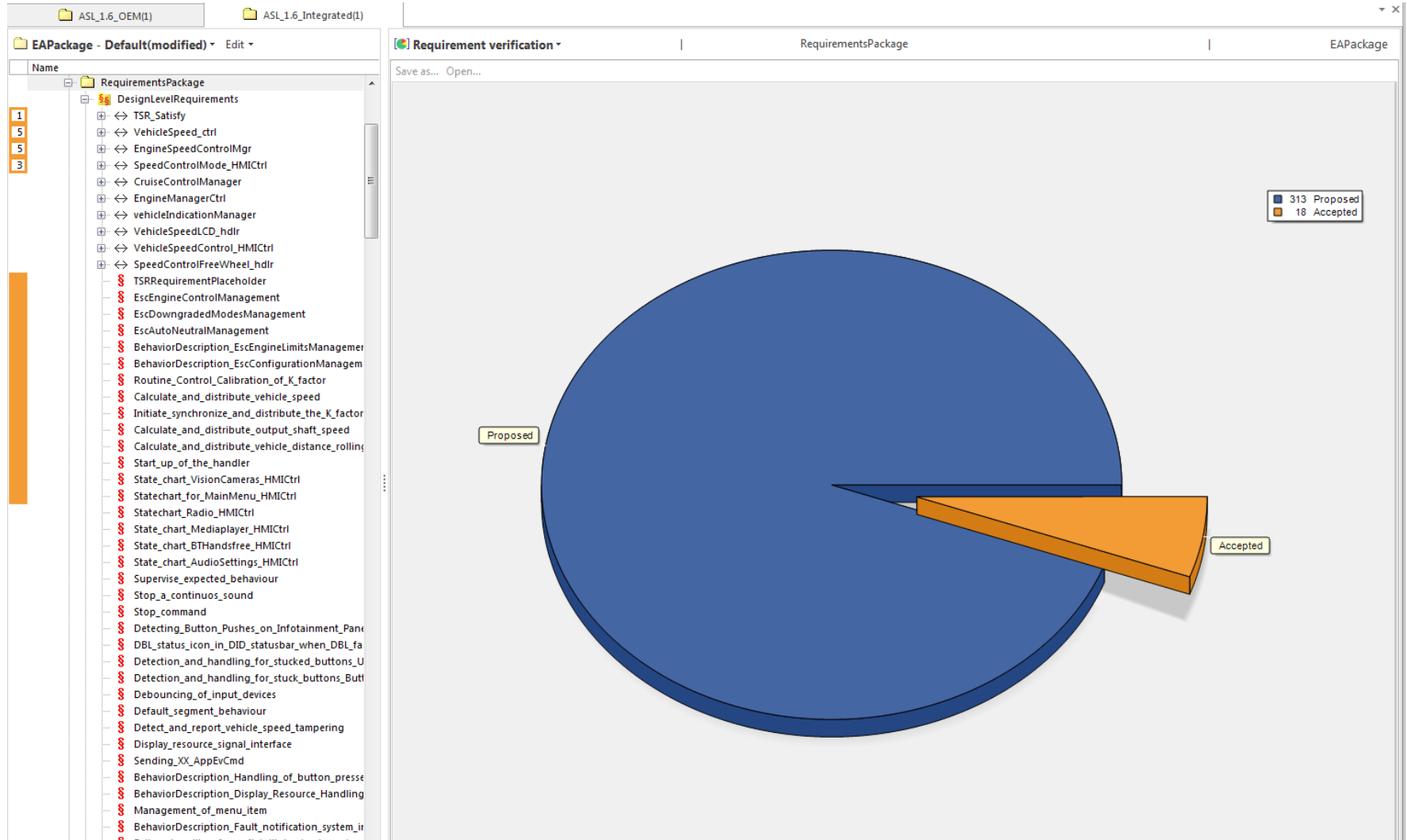
- 8_TSRComponentDependability.sgraphml
 - ASL_1.6_OEM.eaxml
 - ASL_1.6_OEMPrevious.eaxml
 - ASL_1.6_Tier1.eaxml
 - EAXML [EAXML]
 - Extensions [EAPackage]
 - Structure [EAPackage]
 - AnalysisLevelElements [EAPackage]
 - DesignLevelElements [EAPackage]
 - DataTypes [EAPackage]
 - FunctionalDesignElements [EAPackage]
 - ASL [EAPackage]
 - DesignFunctionTypes [17 items]
 - C [DesignFunctionType]
 - LDC_AcceleratorPedalCtrl [DesignFunctionType]
 - LDC_CruiseControlManager [DesignFunctionType]
 - LDC_EngineManagerCtrl [DesignFunctionType]
 - LDC_EngineSpeedControlMgr [DesignFunctionType]
 - LDC_RetarderCtrl [DesignFunctionType]
 - LDC_RoadSpeedLimitManager [DesignFunctionType]
 - LDC_SpeedControlMode_HMICtrl [DesignFunctionType]
 - LDC_VehicleIndicationManager [DesignFunctionType]
 - LDC_VehicleSpeedControl_HMICtrl [DesignFunctionType]
 - LDC_VehicleSpeedLCD_hdlr [DesignFunctionType]
 - S [DesignFunctionType]
 - SpeedControlButtons_2_hdlr [DesignFunctionType]
 - SpeedControlFreeWheel_hdlr [DesignFunctionType]
 - vehicle_model [DesignFunctionType]
 - VehicleSpeed_ctrl [DesignFunctionType]
 - VehicleSpeedControl_HMICtrl [DesignFunctionType]
 - FunctionalDesignArchitecture [DesignFunctionType]
 - TSR [EAPackage]
 - TSR_Internal_Architecture [EAPackage]
 - TSR_Internal_Function_Types [EAPackage]

RequirementsPackage

DesignFunction...	uuid	Satisfied Requirements
LDC_CruiseCon...	x0400000000187F6F	Detecting_Button_Pushes_on_Infotainment_Panel DBL_status_icon_in_DID_statusbar_when_DBL_failure Detection_and_handling_for_stucked_buttons_Updated_Gear_... Detection_and_handling_for_stuck_buttons_Button_panel_ Debouncing_of_input_devices Default_segment_behaviour Detect_and_report_vehicle_speed_tampering
LDC_RoadSpee...	x0400000000187F86	
TSR	x04000000005FBE2A	TSRRequirementPlaceholder TSR_A1 TSR_A2 TSR_A3 TSR_A4

Import

Metrics



Export

